Nitin Singh

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Summary

Staff Machine Learning Engineer at Aurora Innovation, specializing in on-device perception with over 8 years of experience in Machine Learning, Computer Vision. My current work focuses on camera and LiDAR-based 3D object detection models, with emphasis on scaling for high accuracy while leveraging quantization, TensorRT and custom CUDA kernels for optimized on-device deployment.

Education

Carnegie Mellon University— Robotics Institute, *Pittsburgh, PA*
Master of Science in Computer Vision — *Aug. '18 – Dec. '19*

Indian Institute of Technology, Kanpur, *Kanpur, India*
Bachelors of Technology in Electrical Engineering, — *Aug. '11 – Apr. '15*

Work Experience

Aurora Innovation Inc, *Mountain View, CA*
Staff ML Engineer, Onboard Perception, 3D Object Detection — *Jan. '21 – Ongoing.*

- ***Model Design*** : Modernized the perception stack by developing transformer based sparse LiDAR encoders to replace dense convolutions, increasing model efficiency and enabling deployment on lower-cost compute. Designed multi-sensor fusion architectures integrating data from 5 LiDARs and 12 cameras, leveraging calibration and multi sweep synchronization to achieve high precision and recall across diverse environments and traffic scenarios, enhancing operational safety.
- ***Model Deployment*** : Designed and implemented custom CUDA kernels for LiDAR–camera fusion, enabling real-time processing of 12 camera streams every 100 ms and improving long-range pedestrian and cyclist recall by 20%. Enabled INT8 quantization of detection model to reduce latency and improve hardware support for deployment.
- ***Model Planning and Triage***: Owned all vehicle detection improvements to unblock commercial operations in challenging conditions, including night driving, sun glare, and highway merging.
- ***Evaluation Infra*** : Implemented evaluation pipeline to computes metrics directly from extracted tensors instead of stack runs, achieving a 20 times reduction in evaluation cost. Enabled frequent large-scale evaluation during training, improving visibility into in-training performance.

Uber, Advanced Technologies Group(ATG), *San Francisco, CA*
Autonomy Engineer II, Perception-Prediction Group — *Feb. '20 – Jan. 21*

- Implemented joint LiDAR-centric and camera-centric detection architecture instead of two separate inference streams. This enhanced sensor fusion and reduced compute required for inference while maintaining performance.

Samsung Research Institute, *Bangalore, India*
Senior Software Engineer with Camera Solutions Group — *May. '15 – Aug. '18*

- Developed the Bokeh effect for Samsung camera using dual-focus images and a CNN-based segmentation model. Integrated data from both dual-focus and segmentation through graph cuts to achieve high segmentation accuracy at edges, enabling commercialization across all devices worldwide.

Academic Projects

Trajectory Prediction for Self-Driving Cars
Guide: Prof. Jeff Schneider — *Jan. '19 – Dec. '19*
Used prior motion cues, scene features and interactions between agents to predict future trajectory of actors. ***Report.***

Publications **Google scholar**

- Uncertainty-aware Short-term Motion Prediction of Traffic Actors for Autonomous Driving, IEEE Winter Conference on Applications of Computer Vision (WACV '20) ***Paper, Open Access, CVF.***
- Depth Aware Portrait Segmentation using Dual Focus Images. IEEE International Conference on Multimedia and Expo (ICME), 2018. ***IEEE.***
- Saliency Prediction with Deep Multiscale Residual Network. Computer Vision and Image Processing 2018. ***Paper***

Patents

- Method and image capturing device for generating high resolution image of region of interest. ***Reference***
- Electronic device and method for automatic human segmentation in image: ***Reference***

Courses & Technical Skills

Grad Courses: Computer Vision, Machine Learning, Visual Learning and Recognition, Geometry in Vision, Math Fundamental for Robotics, Simultaneous Localization and Mapping

Programming: Python, C++

Frameworks : Pytorch, Tensorflow